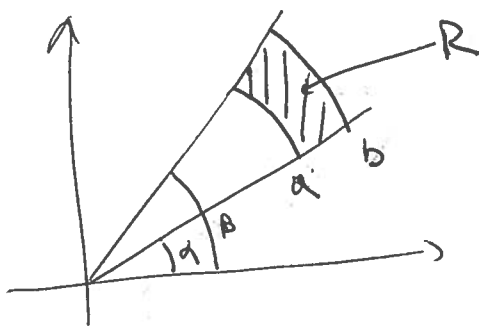


§ 13.4 Double integrals in Polar coords.

A polar rectangle is given by

$$R = \{ (r, \theta) \mid a \leq r \leq b, \alpha \leq \theta \leq \beta \}$$



The area of R is approximately $(b-a) \times R(\beta-\alpha)$. If $(b-a)$ & $(\beta-\alpha)$ are small, this approximation is good, so we let $\Delta A = r \Delta r \Delta \theta$. Then

$$V = \iint_R F(r, \theta) r dr d\theta = \iint_R f(x, y) dA$$

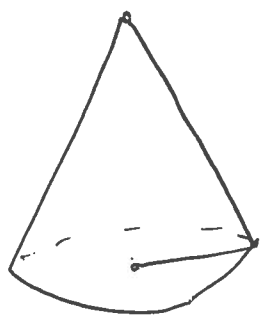
gives the volume of the region over R bounded by the function $z = f(x, y)$.

This is useful when the region has the right symmetries

$$\begin{aligned} \Sigma x \quad R &= \{ 1 \leq r \leq 2, 0 \leq \theta \leq \pi/4 \} & \iint_R \cos(x^2 + y^2) dA &= \int_0^{\pi/4} \int_1^2 \cos(r^2) \cdot r dr d\theta \\ &= \int_0^{\pi/4} \left. \frac{1}{2} \sin(r^2) \right|_1^2 d\theta &= \int_0^{\pi/4} \frac{1}{2} (\sin 4 - \sin 1) d\theta &= \frac{\pi}{8} (\sin 4 - \sin 1) \end{aligned}$$

Ex Volume of a cone of radius R & height H

$$B = \{(r, \theta) \mid 0 \leq r \leq R, 0 \leq \theta \leq 2\pi\}$$



$$z = H - \frac{H}{R} \cdot r$$

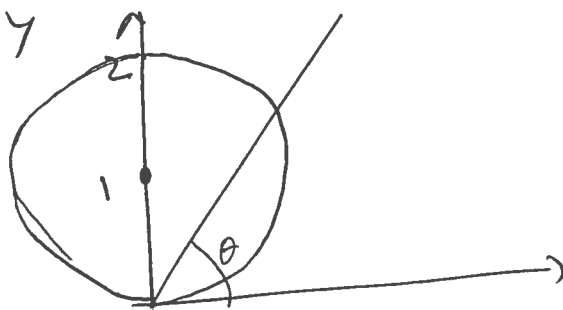
$$\int_0^{2\pi} \int_0^R H \left(1 - \frac{r}{R}\right) r \, dr \, d\theta = \int_0^{2\pi} \left[\frac{Hr^2}{2} - \frac{Hr^3}{3R} \right]_0^R d\theta$$

$$= \int_0^{2\pi} H \frac{R^2}{2} - \frac{HR^2}{3} d\theta = \int_0^{2\pi} \frac{HR^2}{6} d\theta = \frac{2\pi HR^2}{6} = \frac{\pi R^2 \cdot H}{3}$$

Ex Volume of solid under $z = x^2 + y^2$ above $z=0$
inside cylinder $x^2 + y^2 = 2y$

$$x^2 + (y-1)^2 = 1 \quad r^2 = 2r \sin \theta$$

$$r = 2 \sin \theta$$



$$V = \iint_S (x^2 + y^2) \, dA = \int_0^\pi \int_0^{2 \sin \theta} r^2 r \, dr \, d\theta$$

$$= \int_0^\pi \left[\frac{r^4}{4} \right]_0^{2 \sin \theta} d\theta = \int_0^\pi 4 \sin^4 \theta \, d\theta = 2 \int_0^{\pi/2} 4 \sin^4 \theta \, d\theta$$

$$= 8 \frac{3}{8} \frac{\pi}{2} = 3\pi$$

↑
formula in the back of the book!

Σ_x Compute $I = \int_0^\infty e^{-x^2} dx = \lim_{b \rightarrow \infty} \int_0^b e^{-x^2} dx = \lim_{b \rightarrow \infty} I_b$

$$V_b = \int_{-b}^b \int_{-b}^b e^{-x^2-y^2} dy dx = \int_{-b}^b e^{-x^2} \left[\int_{-b}^b e^{-y^2} dy \right] dx = \int_{-b}^b e^{-x^2} \cdot 2I_b dx$$

$$= \int_{-b}^b e^{-x^2} dx \cdot 2I_b = 4I_b^2$$

$V = \lim_{b \rightarrow \infty} V_b = 4I^2$ Volume under $z = e^{-x^2-y^2}$ over $z=0$

$$V_\bullet = \lim_{a \rightarrow \infty} \int_0^{2\pi} \int_0^a e^{-r^2} r dr d\theta = \lim_{a \rightarrow \infty} \int_0^{2\pi} \left[-\frac{1}{2} e^{-r^2} \right]_0^a d\theta$$

$$= \lim_{a \rightarrow \infty} \int_0^{2\pi} \left(-\frac{1}{2} e^{-a^2} + \frac{1}{2} \right) d\theta = \lim_{a \rightarrow \infty} \pi(1 - e^{-a^2}) = \pi$$

So $4I^2 = \pi$ so $I = \frac{\sqrt{\pi}}{2}$

Σ_x Compute a) $\int_0^{\pi/2} \int_0^{\cos \theta} r^2 \sin \theta dr d\theta$, b) $\int_0^{2\pi} \int_0^\theta r dr d\theta$

c) the area inside the circle $r = 4 \cos \theta$ & outside $r = 2$.

d) the area of one leaf of the 4-leaved rose $r = a \sin 2\theta$

e) $\iint_S e^{x^2+y^2} dA$ where $S = \{(x,y) \mid x^2+y^2 \leq 4\}$

f) $\iint_S \sqrt{4-x^2-y^2} dA$ $S = \{(x,y) \mid x^2+y^2 \leq 4, x \geq 0, y \geq 0\}$

g) $\iint_S y dA$ $S = \{(x,y) \mid 1 \leq x^2+y^2 \leq 4, x \geq 0, y \geq 0\}$